Naïl LETTY

nail.letty@gmail.com
+33 7 68 43 62 12
LinkedIn : www.linkedin.com/in/nail-letty
Nationality: French
Engineering student at ISAE-SUPAERO

SUMMARY

1st year of Master of Science ISAE-SUPAERO engineering student (graduation 2027) focused on quantum systems and hardware-adjacent control. Strong foundations in applied physics, optimization (with and without constraints), electronics, and control theory (feedback, compensators), with practical compiling control-ready circuits. Comfortable with Python/Qiskit and MATLAB for simulation and tooling; motivated to contribute to quantum hardware, calibration, and control/benchmarking workflows.

EDUCATION (Highest level of education: Bachelor of Science

Ecole Polytechnique / Coursera: Quantum Optics2: Two Photons and more November 2025

Photo-detection, electrical engineering, entanglement, Quantum technologies

Ecole Polytechnique / Coursera: Quantum Optics1: Single Photons November 2025

Quantum Mecanics, Quantum Optics, Quantization of electromagnetic field, photo-detection, electrical engineering

IBM Certifications August-September 2025

Quantum Machine Learning; Variational Algorithms Design; Quantum Diagonalization Algorithms; Practical Introduction to Quantum-Safe Cryptography

National Higher French Institute of Aeronautics and Space (ISAE-SUPAERO) TOULOUSE, FRANCE 2024 – 2028

(Currently in first year of Master of Science)
Core studies in applied physics, mathematics for optimization (with/without constraints), electronics, signals & systems, and control engineering (PID/correctors), plus quantum mechanics and quantum engineering modules.

PROFESSIONAL EXPERIENCE

Experimental Quantum Optics Practice – ISAE-SUPAERO, Toulouse, France April-June 2025

Generated polarization-entangled photon pairs via SPDC in a BBO crystal pumped by a laser; counted singles and coincidences with silicon APDs; characterized optical throughput and polarization-correlation fringes; prepared a CHSH Bell-parameter measurement to evidence nonclassical entanglement.

PERSONAL EXPERIENCE

Bloch Simulator Project August-October 2025

I developed a Python/Qiskit simulator for visualizing 1- and 2-qubit state evolution, including noise models (T_1, T_2, depolarization) and metrics such as fidelity, purity, and concurrence. It included the visualization of continuous Bloch-sphere trajectories through real-time animations. Open source: github.com/lettynail/bloch_simulator

Quantum Teleportation Study September 2025

I generalized the quantum teleportation circuit to the teleportation of n states put in parallel, compiled the circuit and executed it on a fake backend, returning Bob's qubit distribution of probability for each teleported state with Qiskit. Open source: https://github.com/lettynail/Quantum_Teleportation

LANGUAGES, TECHNOLOGY & OTHER

Quantum: Qiskit, noise models (T_1/T_2/depol), VQE basics, transpilation.
Software: Python (NumPy/SciPy), Git/GitHub, Jupyter.
Other: C, Java, MATLAB, CAD (CATIA, SolidWorks), STIM, Crumble, fundamentals of analog/digital electronics, control concepts (feedback, compensators), PATRAN/NASTRAN.
Languages: French (native), English (C1 – TOEFL).